(Please write your Exam Roll No.)

Exam Roll No. ....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY-FEBRUARY 2023

Paper Code: ICT201 | **Subject: Foundation of Computer Science**

Time: 3 Hours | **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit.

Q1 Attempt **any five:** **(3x5=15)**

a) Define Well ordering principle

7 b) Show $[p \wedge (p \rightarrow q)] \rightarrow q$ is a tautology

3 c) What is the smallest and largest possible equivalence relation on the set A

d) Solve recurrence relation for the tower of hanoi problem

e) what is the best, worst and average case complexity of the quick sort algorithm

3 f) State lagrange's therorem ?

4 g) Is this necessary for a cyclic group to be abelian?

3 h) Prove the uniqueness of identity and inverse elements in a Group.

3 i) Differentiate in between hamilton and Euler's path.

UNIT-I

Q2 Define proof by contradiction and proof by exhausive cases. Suppose x □ Z (set of integers). Prove that If 7x+9 is even, then x is odd using a contrapositive proof. **(15)**

Q3 Define tautology and contradiction? Given the premises (□x)P(x) and (∀x)[P(x) → Q(x)] give a series of steps concluding that (□x)Q(x) **(15)**

UNIT-II

Q4 What is a recursion tree? State and prove Master's theorem. Define the use of regulatory condition. **(15)**

Q5 Find the generating function for the sequence given recusively by $a_n = 2a_{n-1} + 4a_{n-2}$ with $a_0 = 1$ and $a_1 = 3$ **(15)**

UNIT-III

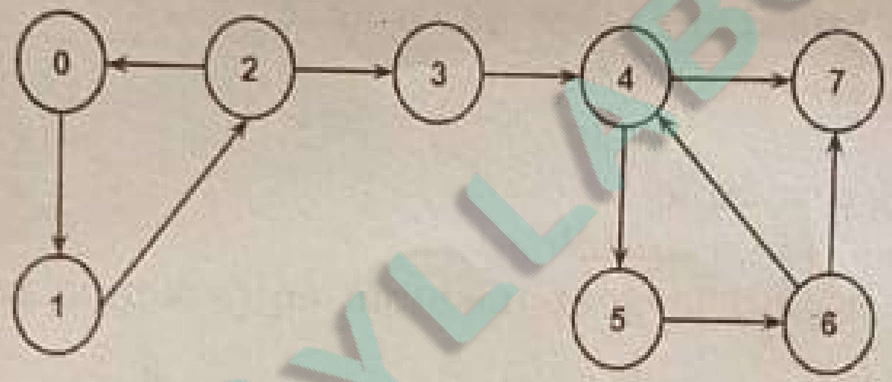
Q6 What is a Normal subgroup? Let G = S_3 be the group of permutations of 3 elements, that is G = {(1),(12),(13),(23),(123),(132)} and let H = {(1),(12)} be a subgroup. Compute the left and right cosets of H. **(15)**

Q7 Define permutation group and cyclic group. Show that [Z12;+12], where +12 is addition modulo 12, is a cyclic group find its generator(s). **(15)**

P.T.O.

UNIT-IV

Q8 Define strongly connected component. Find the strongly conncetd compnents of the following grpah [clearly state pseudocode of the algorithm used] **(15)**



Q9 State and proved Euler's formula for a connected planar simple graph. **(15)**

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